

Thanh-Long V. Le

 Homepage  Google Scholar  LinkedIn  Email

Research Interests

My research centers on **large language models (LLMs)**, with an emphasis on **improving their reasoning abilities through reinforcement learning and other post-training techniques**. Recently, I have also begun exploring generative modeling, focusing on **applying reinforcement learning to diffusion and flow models**.

Education

Korea Advanced Institute of Science and Technology (KAIST AI)

Feb. 2025 – Feb. 2027

M.S. in Artificial Intelligence

Seoul, South Korea

- GPA: **4.10/4.30**
- Graduate Researcher at MLILAB

Korea Advanced Institute of Science and Technology (KAIST)

Sep. 2019 – Feb. 2024

B.S. in Computer Science and Artificial Intelligence

Daejeon, South Korea

- GPA: **3.98/4.30**
- Undergraduate Researcher at MIIL
- Graduated **Magna Cum Laude** (ranked top 8% in School of Computing)
- Minor in Business and Technology Management

Publications

No Prompt Left Behind: Exploiting Zero-Variance Prompts in LLM Reinforcement Learning via Entropy-Guided Advantage Shaping

Thanh-Long V. Le, Myeongho Jeon, Kim Vu, Viet Lai, Eunho Yang

International Conference on Learning Representations (ICLR) 2026

Design Opportunities for Explainable AI Paraphrasing Tools

Yewon Kim, Thanh-Long V. Le, Donghwi Kim, Mina Le, Sung-Ju Lee

Conference on Designing Interactive Systems (DIS) 2025

(FL)²: Overcoming Few Labels in Federated Semi-Supervised Learning

Seungjoo Lee, Thanh-Long V. Le, Jaemin Shin, Sung-Ju Lee

Conference on Neural Information Processing Systems (NeurIPS) 2024

FlowTool: Controlling Tool Parameter In Image Editing Via Flow Matching

Thanh-Long V. Le, Steven Walton, Seunghyun Yoon, Branislav Kveton, Trung Bui, Eunho Yang, Viet Lai

Under review at *International Conference on Learning Representations (ICLR) 2027*

Beyond Scores with Reward-Model Embeddings for Diverse Flow RL

Yunfei Xie, Yidong Huang, Thanh-Long V. Le, Chen Wei, Anup Rao, Viet Lai, Subhojyoti Mukherjee, Branislav Kveton

Under review at *International Conference on Learning Representations (ICLR) 2027*

Research Experience

EPFL

Jan. 2027 – May. 2027

Visiting Researcher, Advised by Prof. Maria Brbić

Lausanne, Switzerland

- Conducting research on AI Agents, World Models, and Video Generation (tentatively).

Adobe Research

Aug. 2025 – Present

Research Scientist Intern, Advised by Dr. Viet Lai

San Jose, CA, USA

- Led the RL-ZVP project, developing a novel algorithm that enhances LLM reinforcement learning by leveraging learning signals from zero-variance prompts.
- Currently working on improving existing RLVR techniques on text-to-image models.

Machine Learning and Intelligence Lab, KAIST AI

Feb. 2025 – Present

Graduate Researcher, Advised by Prof. Eunho Yang

Seoul, South Korea

- Conducting research on LLM post-training, including LLM reasoning and alignment through reinforcement learning/preference optimization.
- Currently focusing on augmenting RLVR with various dense learning signal to improve the reasoning ability of LLM.
- Leading a collaborative project with AITRICS on developing a large reasoning model for highly imbalanced clinical event prediction using patient EHR data.

Mobile Intelligence & Interaction Lab, KAIST

May. 2023 – Sep. 2024

Undergraduate Researcher, Advised by Prof. Sung-Ju Lee

Daejeon, South Korea

- Conducted research on natural language processing, federated learning and human-computer interaction.
- Actively participated in the Exphrase project – an AI-based paraphrasing system equipped with five supplementary features: AI Translation, AI Confidence Score, AI Explanation, Example Usages, Statistics.
- Addressed label deficiency in federated learning by implementing semi-supervised learning with adaptive thresholding and sharpness-aware regularization.

Work Experience

Luxoft – BMW Korea

Mar. 2024 – Dec. 2024

C++ Software Engineer

Seoul, South Korea

- Developed and optimized features for map provision application to enhance the accuracy and reliability of ADAS (Advanced Driving Assistance System) functionalities of BMW vehicles.
- Detected, analyzed, and prepared reports and fixes for vehicle anomalies by leveraging software logs and simulations.

Coc Coc Company Ltd.

Sep. 2021 – Oct. 2022

Software Engineer

Hanoi, Vietnam

- Maintained the functionality of three key components of Coc Coc Search's backend system – compositor, knowledge graph, and search engine.
- Enhanced the quality of Coc Coc's Answer Box by implementing multilayer rule-based question-answering system and incorporating various alternative logic to address common query patterns, resulting in a 200% increase in impression and a 30% increase in click-through rate.
- Integrated Prometheus into services to enable automatic recording of metrics, eliminating the need for manual retrieval via logs.
- Conducted a full codebase refactoring of compositor system to eliminate potential bugs and technical debt.
- Utilized SQL to collect and analyze usage data of Coc Coc Translate over the course of a year, providing insights to assist managers in decision making.

Honors and Awards

Qualcomm-KAIST Innovation Award (2023): Winner.

KAIST Dean's List (2020): Top 3% students of the department.

KAIST International Undergraduate Scholarship (2019): Full-ride scholar.

Deakin Vice-Chancellor's International Scholarship (2019): Recipient.

Vietnam National Mathematical Olympiad (2018): Silver medal.

Technical Skills

Languages: C++, C, Python, Java, Go, SQL, HTML, CSS, JavaScript.

Technologies/Frameworks: PyTorch, verl, TRL, Unsloth, Jax, Haliar, Levanter, Linux, Git, Vim, Bazel, Bash Scripting.

Certificates: 8.0 in IELTS, 1500 in SAT, 800 in SAT Math Level 2, 800 in SAT Physics.

Academic Services

Conference Reviewer: NeurIPS 2026, EMNLP 2026, ICLR 2027.